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MODERN DELAY-TOLERANT EMAIL

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Abstract

Abstract content

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1 | Introduction

1.1 Background

Communication in space is not like communication on Earth. The distances are large, so signals take time to travel: a one-way trip between Earth and the Moon takes about 1.3 seconds, and to Mars it takes minutes. Links also break for long periods, as bodies move and block the line of sight. The normal internet assumes the opposite, short delays and a connection that stays up, so its protocols fail here: they expect a reply within a short time and treat a missing one as an error.

Delay-Tolerant Networking (DTN) was designed for these conditions (1). Instead of a live end-to-end connection, it uses store-and-forward: each node holds a message until the next link is available, then passes it on. The Bundle Protocol carries messages as self-contained units called bundles between DTN nodes (2). This tolerates long delays and broken links, which makes DTN the accepted basis for interplanetary networking (3), an active area with real implementations (4, 5).

Email is one of the basic tools people expect, so carrying it over DTN has received attention. A gateway translates between standard mail protocols and the Bundle Protocol (6, 7), and open tools implement this (8), so a message can move from a mail server on one side of a DTN link to a mail server on the other.

On any mail system, much of the incoming mail is spam, and anti-spam tools are a standard part of the setup. These tools do more than read the message text. They also test the sender and the links in the message against live network services: DNS

blocklists of known-bad addresses, sender reputation systems, and shared databases of known spam. Every one of these checks sends a request to a remote server during the scan and waits for a fast answer before it can score that rule.

1.2 Motivation

Work on DTN email has gone into moving mail across the link, not into filtering it once it arrives. Yet filtering is part of running a mail service, not an optional extra. Consider a crewed Moon base whose users exchange email with Earth over a DTN link (9): spam will reach it as it reaches any mail system, and the tools that would catch it now depend on lookups that must cross the delayed link.

Whether those tools still work is not obvious. A DTN link breaks the assumption they rely on: the answer is now seconds away, or the link is down when the request is made. A tool might slow to a crawl while it waits, lose detection when answers arrive too late, give up early and let spam through, retry until it loads the link, or fail with no sign that it failed. Each tool reaches the network in its own way, so each may respond differently. None of this has been measured.

This dissertation measures it. The value is in understanding how the tools behave under delay and why, which is a starting point for anyone who later builds spam filtering into interplanetary email.

1.3 Research Question and Objectives

This dissertation turns that gap into a precise question.

Research question. The primary research question is:

How do existing anti-spam tools behave when run over a delay-tolerant network link, and what mechanisms drive that behaviour?

It breaks into three sub-questions:

- How does each tool's network dependence interact with propagation delay?
- What failure modes appear under delay, and what causes them?
- How do the tools' own timeout and retry settings behave under delay, and does tuning them change the outcome?

Objectives. To answer this, the dissertation sets out to:

1. Build a controlled Earth–Moon DTN email testbed using real mail and DTN software, with a link delay that can be set to a chosen value.
2. Assemble a representative email corpus and a repeatable method to scan it and record per-email timing, verdicts, and network traffic.
3. Evaluate four tools, SpamAssassin, Rspamd, Vipul's Razor, and DCC, chosen to cover the main ways a filter depends on the network.
4. For each tool, compare its behaviour under delay against a no-delay baseline and identify the mechanisms behind the differences and any failure modes.
5. Draw the cross-tool comparison: what generalises about anti-spam filtering that depends on synchronous network services when it runs under delay.

Scope. This is a behavioural study: it characterises and explains the tools' response to delay, and does not propose or build a filtering system for interplanetary email.

Contributions. The dissertation makes three contributions:

- A reproducible Earth–Moon DTN email testbed for evaluating mail-layer tools under controlled delay.
- An empirical, mechanism-level account of how four anti-spam tools behave under delay, including the failure modes and their causes.
- A measurement method that separates sourced fact from interpretation,

controls for blacklist drift, and uses back-to-back runs so that delay effects can be isolated.

1.4 Dissertation Overview

This dissertation is organised into six chapters. Chapter 1 has introduced the problem, the research question, and the objectives. Chapter 2 reviews the background: delay-tolerant networking and the Bundle Protocol, the email system and how mail is carried over DTN, and the anti-spam tools and the network services they depend on. Chapter 3 describes the design of the testbed, including the node layout, the choice of software, and the delay simulation. Chapter 4 covers the implementation, from building the nodes to configuring the mail, DTN, and anti-spam software. Chapter 5 presents the evaluation: each of the four tools is measured under delay against a no-delay baseline, and the results are compared. Chapter 6 concludes and sets out directions for future work.

2 | Literature Review

3 | System Design

This chapter describes the testbed used to evaluate anti-spam tools under the one-way delay of a deep-space DTN link. It sets out the node layout, the host model and the reasons for it, the tools under test, the DTN configuration, the delay simulation, and the experimental design.

3.1 System Architecture

The testbed has two networks joined by a relay. Each node is a separate virtual machine running Ubuntu 22.04 LTS. Earth uses the 192.168.100.0/24 subnet and Moon uses the 192.168.200.0/24 subnet. The nodes are:

- **Mail1 (Earth).** The Earth mail server. Runs Postfix, Dovecot, and the four anti-spam tools. Address 192.168.100.10. ION node 1.
- **Space (Relay).** The transit node between the two networks. Runs an ION node and carries the delay and packet-capture configuration. Earth-side interface 192.168.100.1, Moon-side interface 192.168.200.1. ION node 2.
- **Mail2 (Moon).** The Moon mail server. Runs Postfix and Dovecot. Address 192.168.200.11. ION node 3.
- **Client1 (Earth).** User terminal on the Earth network. Runs Thunderbird and connects to Mail1 over IMAP (port 143) and SMTP submission (port 587, STARTTLS).

- **Client2 (Moon).** User terminal on the Moon network. Runs Thunderbird and connects to Mail2 the same way.

Mail moves server to server across the DTN link: a message submitted on one mail server is carried as a bundle through the relay to the other. ION runs on all three servers, so the bundle path is node 1 (Earth) to node 2 (relay) to node 3 (Moon). The relay is the only path between the two networks, so all traffic between Earth and Moon passes through it. This is where the one-way link delay is applied, standing in for the Earth–Moon propagation time. Mailboxes use the mbox format, and each user reads mail from their local server. The topology is shown in Figure 3.1.

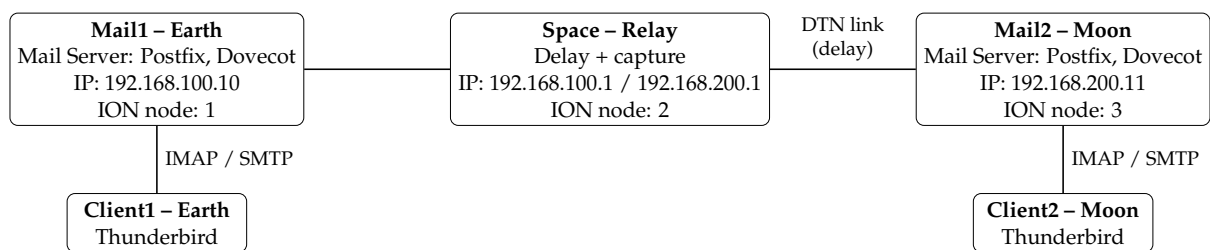


Figure 3.1: Testbed topology. Each client connects only to its local mail server. The relay is the only path between the Earth and Moon networks and is where the one-way link delay is applied.

Data flow (Earth to Moon). The path of a message is shown in Figure 3.2.

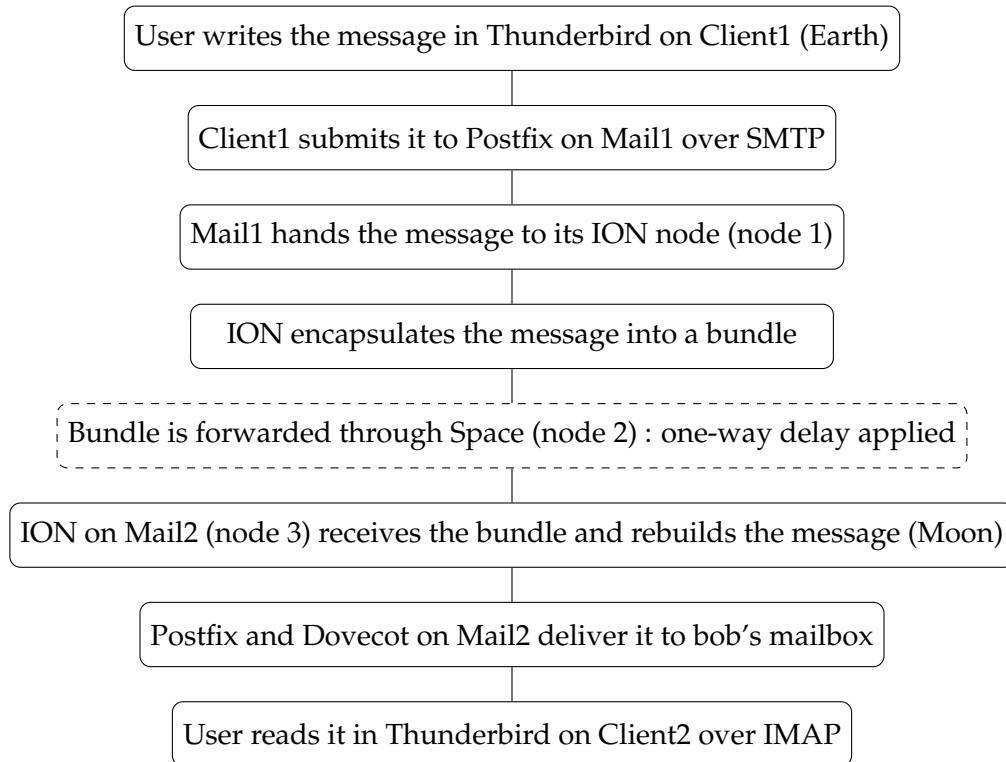


Figure 3.2: End-to-end path of an email from Earth to Moon across the DTN link.

3.2 Environment and Technology Selection

3.2.1 Virtual machines over containers

The testbed runs on full virtual machines (Ubuntu 22.04 LTS, ARM64, under UTM on macOS). Containers were considered but not used. The anti-spam experiment has three needs that decide this:

- **VMs match the real topology.** The testbed mimics three separate machines — Earth, Space, Moon — connected by a long-delay link, and full VMs naturally model this, whereas containers all share one kernel on one host.
- **Simpler delay setup** Setting up `ifb` to delay incoming traffic is fiddly and error-prone inside containers, but it's straightforward on a VM where each node has its own kernel and network stack.
- **DTN state stays isolated per node.** Each ION node keeps its own bundle store

and contact schedule, and a full VM ensures the link delay sits on the path itself rather than leaking into a node's own processing.

3.3 Email stack

- **Postfix** was chosen for its clear interface for handing mail to an external program, which is needed to pass messages to the DTN layer.
- **Dovecot** was chosen because it works directly with the mbox format and the local system accounts that Postfix delivers to.
- **Thunderbird** was chosen as a real mail client, so the message traffic in the testbed matches what a normal user would generate.

3.4 DTN stack

- **ION-DTN** was chosen for its deep-space flight heritage and its existing tool for carrying email over the Bundle Protocol, which connects directly to the mail stack.

3.5 Anti-Spam Tool Selection

Four tools were selected, each with a different external dependency:

- **SpamAssassin 3.4.6** — content scoring that depends on DNS blocklist lookups.
- **Rspamd 2.7** — content scoring with a large number of asynchronous DNS lookups.
- **Vipul's Razor 2.84** — collaborative filtering through a central nonce server, reached over TCP on port 2703.
- **DCC 2.3.169** — collaborative filtering through checksum servers, reached over UDP on port 6277.

The four cover the main ways a filter reaches the network: blocklist DNS (SpamAssassin, Rspamd), a TCP request to a central reputation server (Razor), and a UDP request to a checksum server (DCC). Testing one tool would show only one failure path.

3.6 DTN Design

The DTN layer uses ION-DTN 4.1.4 and BPMail. The three servers run ION as nodes on a single ipn scheme: Mail1 (Earth) is node 1, Space (relay) is node 2, and Mail2 (Moon) is node 3. Bundles are carried over a UDP convergence layer on port 4556, and the relay forwards them between the two mail nodes, so the path from Earth to Moon is node 1 to node 2 to node 3. The contact plan sets the windows during which the nodes can exchange bundles. BPMail sits above ION on each mail server: it takes a message from Postfix, encapsulates it into a bundle for the destination node, and rebuilds it on arrival. The full configuration, including the SDR settings, the contact plan, and the startup order, is given in the implementation chapter.

3.7 Delay Simulation Design

Delay is applied with netem. The key requirement is that delay is applied to one direction only — to traffic coming from the remote node — and not to the Earth node's own traffic.

A plain netem rule on the relay's interface delays everything that crosses it, including the Earth node's traffic, which would change what is being measured. To avoid this, ingress traffic from the remote side is redirected to an Intermediate Functional Block (ifb) device, and the delay is applied there. The redirect happens before the relay's NAT rewrites the source address, so the rule can match traffic by its true origin. Earth-origin traffic does not pass through this path and stays at normal latency. This is shown in Figure 3.3.

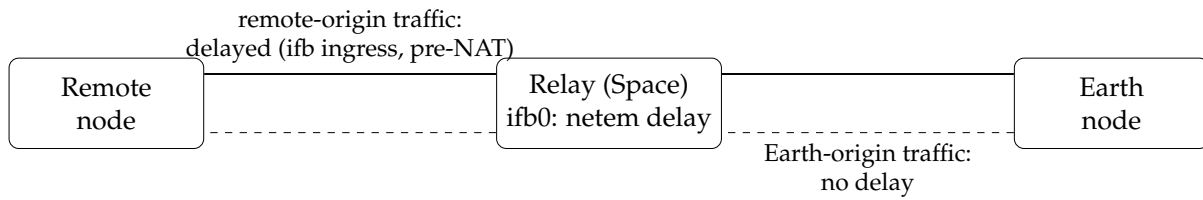


Figure 3.3: Asymmetric delay. Traffic originating from the remote node is delayed on ingress through an ifb device, before NAT rewrites the source. Earth-origin traffic is not delayed.

Delay is the independent variable. The experiments use a representative one-way delay of 1.3 seconds as a deep-space operating point. The testbed allows any value to be set, so the same setup covers the range from short Earth–Moon delays to multi-minute deep-space delays.

3.8 Summary

This chapter set out a three-node testbed for measuring anti-spam tool behaviour under link delay. The design uses full virtual machines for kernel and DNS isolation, a private resolver to give the Earth side a working DNS path, four tools chosen to cover the main external dependencies, and an ifb-based scheme to apply delay to one direction only. The next chapter describes how this design was built and configured, and the chapter after it presents the results.

4 | Implementation

Implementation content

5 | Results and Discussion

Results

6 | Conclusions and Future Work

Conclusions

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